

# CRISTIAN MEZA

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## PERSONAL DETAILS

Date and Place of birth: 30/04/1974. Quilpue, CHILE.  
RUT: 12.602.837-7.  
Chilean, Married.

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## ADDRESS

Instituto de Ingeniería Matemática-CIMFAV  
Facultad de Ingeniería, Universidad de Valparaíso.  
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Email: [cristian.meza@uv.cl](mailto:cristian.meza@uv.cl)  
Web page: <https://sites.google.com/uv.cl/cmeza>  
ORCID: <https://orcid.org/0000-0001-6332-6137>

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## EDUCATION

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| 2003–2006 | <b>Ph.D. in Statistics</b> , Université Paris-Sud, Orsay, FRANCE.<br>Ph.D. Thesis: “Extensions et applications de l’algorithme SAEM pour les modèles mixtes”.<br>Advisers: Marc Lavielle, Université Paris 5 and Jean-Louis Foulley, INRA Jouy-en-Josas. |
| 2002–2003 | <b>M.S. (D.E.A.) “Modélisation Stochastique et Statistique”</b> , Université Paris-Sud, FRANCE.                                                                                                                                                          |
| 2000–2001 | <b>M.S. in Statistics</b> , Pontificia Universidad Católica de Chile, Santiago, CHILE.                                                                                                                                                                   |
| 1995–1999 | <b>B.S. in Mathematics and Statistics</b> , Pontificia Universidad Católica de Chile, Santiago, CHILE.                                                                                                                                                   |

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## RESEARCH INTEREST

Mixed-effects models, REML estimation, EM/SAEM Algorithm, MCMC methods, semiparametric models, Segmentation Models.

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## PUBLICATIONS

John Barrera, **Cristian Meza**, Ana Arribas Gil. “A stochastic method to estimate a zero-inflated two-part mixed model for human microbiome data” . Accepted in Statistical Modelling, 2026.

Danilo Alvares, **Cristian Meza** and Rolando De la Cruz. Bayesian inference for nonlinear mixed-effects location scale and interval-censoring cure-survival models: an application to pregnancy miscarriage. Statistical Methods in Medical Research, 34 (8), 1525–1533, 2025.

Rolando de la Cruz, Marc Lavielle, **Cristian Meza** and Vicente Núñez-Antón. A joint analysis proposal of nonlinear longitudinal and time-to-event right-, interval-censored data for modeling pregnancy miscarriage. (2024) Computers in Biology and Medicine, 182, November 2024, 109186.

Maritza Márquez, **Cristian Meza**, Dae-Jin Lee and Rolando de la Cruz. Classification in semiparametric nonlinear mixed models using P-Splines and the SAEM algorithm. (2023) Statistics in Medicine, 42(27), 4952-4971.

Lucas Vicuna, Esteban Barrientos, Tomas Norambuena, Danilo Alvares, Juan Cristobal Gana, Valeria Leiva-Yamaguchi, **Cristian Meza**, Jose L. Santos, Veronica Mericq, Ana Pereira and Susana Eyheramendy, New insights from GWAS on BMI-related growth traits in a longitudinal cohort of admixed children with Native American and European ancestry. (2023) iScience, 26(2), 106091.

Rolando de la Cruz, **Cristian Meza**, Nicolás Narria and Claudio Fuentes (2022). “A Bayesian Change Point Analysis of the USD/CLP Series in Chile from 2018 to 2020: Understanding the Impact of Social Protests and the COVID-19 Pandemic”. Mathematics, 10(18), 3380.

Rolando de la Cruz, Hugo Salinas, and **Cristian Meza** (2022). “Reliability Estimation for Stress-Strength Model Based on Unit-Half-Normal Distribution”. Symmetry, 14(4), 837.

Gabriel A. Caceres, Eric D. Feigelson, G. Jogesh Babu, Natalia Bahamonde, Alejandra Christen, Karine Bertin, **Cristian Meza** and Michel Curé (2019). “Autoregressive Planet Search: Methodology”. The Astronomical Journal, 158 (2), 57.

Gabriel A. Caceres, Eric D. Feigelson, G. Jogesh Babu, Natalia Bahamonde, Alejandra Christen, Karine Bertin, **Cristian Meza** and Michel Curé (2019). “Autoregressive Planet Search: Application to the Kepler Mission”. The Astronomical Journal, 158 (2), 58.

Meili, Baragatti, Karine Bertin, **Cristian Meza** and Emilie Lebarbier (2019). “A Bayesian approach for the segmentation of series with a functional effect”. Statistical Modelling, 19 (2), pp. 194-220.

Rolando de la Cruz, Claudio Fuentes, **Cristian Meza** and Vicente Núñez-Antón (2018). “Error-Rate Estimation in Discriminant Analysis of Non-Linear Longitudinal Data: A Comparison of Resampling Methods”. Statistical Methods in Medical Research, 27 (4), pp. 1153–1167.

Rolando de la Cruz, Claudio Fuentes, **Cristian Meza**, Dae-Jin Lee and Ana Arribas-Gil (2017). “Predicting pregnancy outcomes using longitudinal information: a penalized splines mixed-effects model approach”. Statistics in Medicine 36 (13), pp. 2120–2134.

- Karine Bertin, Xavier Collilieux Emilie Lebarbier and **Cristian Meza** (2017). “Semi-parametric segmentation of multiple series using a DP-Lasso strategy”. *Journal of Statistical Computation and Simulation*, Volume 87, pp. 1255-1268.
- Rolando de la Cruz, **Cristian Meza**, Ana Arribas-Gil and Raymond Carroll (2016). “Bayesian Regression Analysis of Data with Random Effects Covariates from Nonlinear Longitudinal Measurements”. *Journal of Multivariate Analysis*, Volume 143, pp. 94-106.
- Ana Arribas-Gil, Rolando de la Cruz, Emilie Lebarbier and **Cristian Meza** (2015). “Classification of longitudinal data through a semiparametric mixed-effects model based on Lasso-type estimators”. *Biometrics*, Volume 71 (2), pp. 333-343.
- Ana Arribas-Gil, Karine Bertin, **Cristian Meza** and Vincent Rivoirard (2014). “Lasso-type estimators for Semiparametric Nonlinear Mixed-Effects Models Estimation”. *Statistics and Computing*, vol. 24 (3), pp. 443-460.
- Cristian Meza**, Felipe Osorio and Rolando de la Cruz (2012). “Estimation in nonlinear mixed effects models using heavy-tailed distributions”. *Statistics and computing*, vol. 22 (1), pp. 121-139.
- Cristian Meza**, Florence Jaffrézic and Jean-Louis Foulley (2009). “Estimation in the probit normal model for binary outcomes using the SAEM algorithm”. *Computational Statistics and Data Analysis*, vol. 53 (4), pp. 1350-1360.
- Cristian Meza**, Florence Jaffrézic and Jean-Louis Foulley (2007). “REML estimation of variance parameters in nonlinear mixed effects models using SAEM algorithm”. *The Biometrical Journal*, vol. 49 (6), pp. 876-888.
- Marc Lavielle and **Cristian Meza** (2007). “A parameter expansion version of the SAEM algorithm”. *Statistics and Computing*, vol. 17 (2), pp. 121-130.
- Florence Jaffrézic, **Cristian Meza**, Marc Lavielle and Jean-Louis Foulley (2006). “Genetic analysis of growth curves using the SAEM algorithm”. *Genetic Selection Evolution*, vol. 38, pp. 583–600.

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## PAPERS FOR CONFERENCES

- Céline Hosteins, Marta Avalos, Diego Kauer, **Cristian Meza**, Laurence Delhaes and Raphael Enaud. “Machine Learning for Compositional Microbiome Data to Predict a Clinical Outcome: Are Interpretable Methods Up to the Task?”. 2025 15th IEEE International Conference on Pattern Recognition Systems (ICPRS), Valparaiso, Viña del Mar, Chile, 2025, pp. 1-7, doi: 10.1109/ICPRS66293.2025.11302842.
- Marta Avalos, John Barrera, **Cristian Meza**, Susana Eyheramendy, Raphael Enaud, Louise-Eva Vandenberght, and Laurence Delhaes. “Indoor Microbiome in Asthma: A Differential Abundance Framework Accounting for Zero-Inflation, Compositionality, Spatial Clustering, and Covariate Adjustment.” *Proceedings of the International Conference on Computational Intelligence Methods for Bioinformatics and Biostatistics 2025*.
- Rolando De la Cruz, Susana Eyheramendy, **Cristian Meza** and Felipe Osorio. “Exact estimation procedures in a spatial mixed-effects probit model with binary outcomes”. 2010 Joint Statistical Meetings Proceedings, Section on Statistical Computing, pp. 3629–3638, August 2010.

**Cristian Meza**, Florence Jaffrézic and Jean-Louis Foulley. “Estimation dans un modèle mixte probit normal pour données binaires en utilisant l’algorithme SAEM et ses dérivées”. Actes de conférence des XXXVII Journées de Statistique de la SFDS, June 2007.

**Cristian Meza**. “Extensions de l’algorithme SAEM : version PX et estimation REML”. Actes de conférence des XXXIX Journées de Statistique de la SFDS, June 2005.

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## PARTICIPATION IN RESEARCH PROJECTS

|           |                                                                                                                                                                                                                                           |
|-----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2025-2028 | Associated Researcher in the Associate Team INRIA VALPO, FRANCE.                                                                                                                                                                          |
| 2024-2026 | Principal Researcher in the project: <i>Statistical modeling nonparametric inference and model selection for complex data-SMILE</i> . MATH AMSUD AM-SUD230032, CHILE.                                                                     |
| 2020-2024 | Associated Researcher in the project: <i>Innovative genome-wide association methods for a longitudinal Chilean cohort: with an application to Height, BMI and pubertal age</i> . FONDECYT 1200146, CHILE.                                 |
| 2020-2023 | Associated Researcher in the project: <i>Statistical modeling for novel biochemical mechanisms to target neurological conditions</i> . PCI-CONICYT REDES190038, CHILE.                                                                    |
| 2019–2023 | Principal Researcher in the project: <i>Time-dependent data: efficient estimation procedures in complex semiparametric mixed-effects models and segmentation methods for multiple series with disturbances</i> . FONDECYT 1190801, CHILE. |
| 2018–2019 | Principal Researcher in the project: <i>Statistical and Stochastic modeling for longitudinal data and application in biometrics using semi parametric approaches</i> . PCI-CONICYT REDES170123, CHILE.                                    |
| 2018–2019 | Principal Researcher in the project: <i>SaSMoTiDep-Statistical and Stochastic modeling for time-dependent data</i> . MATHAMSUD-CONICYT 18-MATH-07, CHILE.                                                                                 |
| 2018      | Associated Researcher in the project: <i>Estimación Eficiente Basada en la Norma L1 para modelos Mixtos</i> . PAI-CONICYT MEC Num. 80170061, CHILE.                                                                                       |
| 2016–2017 | Associated Researcher in the project: <i>Statistical inference for dependent stochastic processes and application in renewable energy</i> . MATHAMSUD Num. 16-MATH-03 SIDRE, CHILE.                                                       |
| 2014–2018 | Principal Researcher in the project: <i>New challenges in mixed-effects models</i> . FONDECYT 1141256, CHILE.                                                                                                                             |
| 2014–2018 | Associated Researcher in the project: <i>Adaptative estimation and oracle inequalities</i> . FONDECYT 1141258, CHILE.                                                                                                                     |
| 2013–2015 | Principal Researcher in the project: <i>Tests with dependent functional data</i> . Posdoctorado FONDECYT 3140602, CHILE.                                                                                                                  |

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| 2013–2014  | Principal Researcher in the project: <i>Support to the development of Astro-Statistics in Valparaíso</i> . Fondo Gemini-Conicyt 32120025, 2012, CHILE.                                                                                                                      |
| 2013       | Associated Researcher in the project: <i>Fortalecimiento de la Astroestadística en Valparaíso</i> . Programa Apoyo a la Formación de Redes Internacionales entre Centros de Investigación, Conicyt, 2012, CHILE.                                                            |
| 2012–2015  | Associated Researcher in the project: <i>Red de Análisis Estocástico y Aplicaciones (sistemas abiertos, energía y dinámica de la información)</i> . Cuarto Concurso Nacional de Anillos de Investigación en Ciencia y Tecnología 2011, CHILE.                               |
| 2009–2013  | Principal Researcher in the project: <i>Exact estimation procedures in mixed-effects models</i> . FONDECYT 11090024, CHILE.                                                                                                                                                 |
| 2009–2011  | Associated Researcher in the project: <i>Desarrollo de un modelo de gestión y diseño transdisciplinario para la vigencia de la unidad de barrio</i> . DIPUV No. 38/2008, Universidad de Valparaíso, CHILE.                                                                  |
| 2008–2010  | Principal Researcher in the project: <i>Métodos de estimación en modelos con efectos mixtos para datos no normales</i> . DIPUV No. 05/2007, Universidad de Valparaíso, CHILE.                                                                                               |
| 2007–2012  | Principal Researcher in the project: <i>Inserción de académicos en la Facultad de Ciencias de la Universidad de Valparaíso, con perspectivas a la creación de un programa de doctorado en ciencias mención modelación estocástica</i> . CONICYT PBCT project PSD 20, CHILE. |
| 2006–2008  | Associated Researcher in the project: <i>Estimation et sélection de modèles dans les modèles non linéaires à effets mixtes et applications en biostatistiques</i> . ANR MONOLIX, FRANCE.                                                                                    |
| Desde 2007 | Associated Researcher in the project: <i>Centro de Investigación y Modelamiento de Fenómenos Aleatorios - Valparaíso, (CIMFAV)</i> . CID 03/2007. Dirección de Investigación. Universidad de Valparaíso, CHILE.                                                             |

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## PROFESSIONAL CAREER & TEACHING EXPERIENCE

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|------------|------------------------------------------------------------------------------------------------------------------------------------------|
| Since 2024 | Full Professor, INGEMAT–CIMFAV, Facultad de Ingeniería, Universidad de Valparaíso, CHILE.                                                |
| 2012–2024  | Associate Professor, CIMFAV–Facultad de Ingeniería, Universidad de Valparaíso, CHILE.                                                    |
| 2007–2011  | Assistant Professor, Departamento de Estadística, Universidad de Valparaíso, CHILE.                                                      |
| 2006–2007  | Teaching Assistant (A.T.E.R.) for graduate and undergraduate courses, Laboratoire de Mathématiques, Université Paris Sud, Orsay, FRANCE. |

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| 2002 | Professor EPES program, Ministry of Education of Chile, Santiago, CHILE.<br>Course: Statistics.                                                                                |
| 2002 | Professor at the Department of Mathematics and Statistics, Pontificia Universidad Católica de Chile. Santiago, CHILE.<br>Course: Statistics and Computing, M.S. in Statistics. |

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## SUPERVISION

Ph.D. in Mathematics:

- Freddy López, UV-PUCV-UTFSM, 2017;
- Maritza Márquez, UV-PUCV-UTFSM, 2022;
- John Barrera, UV-PUCV-UTFSM, 2025.

M.S. in Statistics: 11 students.

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## INSTITUTIONAL RESPONSIBILITIES

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| 2023-2025 | “Presidente del Consorcio” of Ph.D. in Mathematics program, the Universidad de Valparaíso-Pontificia Universidad Católica de Valparaíso-Universidad Santa María. |
| 2023-204  | Member of “Grupo de Estudio Matemáticas” Postodoc and Iniciación FONDECYT-ANID.                                                                                  |
| 2012-2016 | Director of Ph.D. in Mathematics program in the Universidad de Valparaíso.                                                                                       |
| 2012-2015 | Member of the “Strategic Issues Committee” of the International Biometrics Society (IBS).                                                                        |
| 2011-2012 | Coordinator of the Chilean Region of the IBS.                                                                                                                    |

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## VISITING

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|------|-----------------------------------------------------------------------------------------------------------|
| 2025 | Visiting Professor (December)<br>Instituto de Matemática, Universidade Federal do Rio de Janeiro, BRAZIL. |
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| 2025 | Visiting Professor (July)<br>Institut de Santé Publique, d'Epidémiologie et de Développement, Université<br>de Bordeaux, Bordeaux, FRANCE. |
| 2025 | Visiting Professor (June)<br>Modal'X, Université Paris Nanterre, FRANCE                                                                    |
| 2024 | Visiting Professor (December)<br>Modal'X, Université Paris Nanterre, FRANCE                                                                |
| 2024 | Visiting Professor (December)<br>Universidad Carlos III de Madrid, Madrid, SPAIN.                                                          |
| 2024 | Visiting Professor (March)<br>Institut de Santé Publique, d'Epidémiologie et de Développement Université de<br>Bordeaux, Bordeaux, FRANCE. |
| 2023 | Visiting Professor (June)<br>Universidad Carlos III de Madrid, Madrid, SPAIN.                                                              |
| 2022 | Visiting Professor (November)<br>Oregon State University, Corvallis, USA.                                                                  |
| 2022 | Visiting Professor (October)<br>CUNY-School of Medicine, New York City, USA.                                                               |
| 2022 | Visiting Professor (June)<br>BCAM, Bilbao, SPAIN.                                                                                          |
| 2022 | Visiting Professor (May)<br>Universidad Carlos III de Madrid, Madrid, SPAIN.                                                               |
| 2021 | Visiting Professor (October)<br>Universidad Carlos III de Madrid, Madrid, SPAIN.                                                           |
| 2019 | Visiting Professor (October)<br>CMAP-Ecole Polytechnique, Palaiseau, FRANCE.                                                               |
| 2019 | Visiting Professor (October)<br>CMAP-Ecole Polytechnique, Palaiseau, FRANCE.                                                               |
| 2019 | Visiting Professor (September)<br>Montpellier SupAgro-INRA, Montpellier, FRANCE.                                                           |
| 2019 | Visiting Professor (August)<br>BCAM, Bilbao, SPAIN.                                                                                        |
| 2018 | Visiting Professor (September)<br>CMAP-Ecole Polytechnique, Palaiseau, FRANCE.                                                             |
| 2018 | Visiting Professor (September)<br>Montpellier SupAgro-INRA, Montpellier, FRANCE.                                                           |

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| 2016 | Visiting Professor (September)<br>Institut de Santé Publique, d'Epidémiologie et de Développement Université de<br>Bordeaux, Bordeaux, FRANCE. |
| 2016 | Visiting Professor (July)<br>Departamento de Estadística, Universidad Carlos III, Madrid, SPAIN.                                               |
| 2016 | Visiting Professor (July)<br>AgroParisTech, Paris, FRANCE.                                                                                     |
| 2015 | Visiting Professor (July)<br>Departamento de Estadística, Universidad Carlos III, Madrid, SPAIN.                                               |
| 2014 | Visiting Professor (July)<br>Departamento de Estadística, Universidad Carlos III, Madrid, SPAIN.                                               |
| 2013 | Visiting Professor (July)<br>Departamento de Estadística, Universidad Carlos III, Madrid, SPAIN.                                               |
| 2013 | Visiting Professor(July)<br>ENSAI, Rennes, FRANCE.                                                                                             |
| 2012 | Visiting Professor(April)<br>Departamento de Estadística, Universidad Carlos III, Madrid, SPAIN.                                               |
| 2010 | Visiting Professor (May)<br>Departamento de Estadística, Universidad Carlos III, Madrid, SPAIN.                                                |
| 2009 | Visiting Professor (June)<br>Departamento de Estadística, Universidad Carlos III, Madrid, SPAIN.                                               |
| 2009 | Visiting Professor (May)<br>Departamento de Estadística, Universidad Estatal de Campinas, Campinas,<br>BRAZIL.                                 |
| 2008 | Visiting Professor (July)<br>Quantitative and Applied Genetics department, INRA, Jouy-en-Josas, FRANCE.                                        |
| 2007 | Visiting Professor (September)<br>Quantitative and Applied Genetics department, INRA, Jouy-en-Josas, FRANCE.                                   |

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## PRESENTATIONS

Séminaire Biostatistique ISPED-Université de Bordeaux, March 2024, Bordeaux, France: “Estimation of a two-part mixed effects model for longitudinal compositional data using the SAEM algorithm”.



XIX Conferencia Española-VIII Encuentro Iberoamericano de Biometría, June 2023, Vigo, Spain: “Classification using a joint model of longitudinal data and binary outcomes based on the SAEM algorithm”.

Un encuentro: Congreso en honor a Chichi León y Marc Lavielle, December 2022, Montevideo, Uruguay: “My PhD Thesis with Marc and what followed”.

XIV Congreso Latinoamericano de Sociedades de Estadística, October 2021, Montevideo, Uruguay (online): “Alternative subgroup joint analysis proposal of nonlinear longitudinal and time-to-event data for modeling pregnancy miscarriage”.

XVII Conferencia Española de Biometría, June 2019, Valencia, Spain: “The segmentation of series with a functional effect: a Bayesian approach”.

IV Jornada de Probabilidad y Procesos Estocásticos, December 2018, Bogota, Colombia: “A Bayesian approach for the segmentation of series with a functional effect”.

XXIXth International Biometric Conference, July 2018, Barcelona, Spain: “A Bayesian approach for the segmentation of series with a functional effect”.

XVI Conferencia Española de Biometría, September 2017, Sevilla, Spain: “Predicting Pregnancy Outcomes Using Longitudinal Information: A P-Splines Mixed-Effects Model Approach”.

Joint Statistical Meeting 2017, August 2017, Baltimore, EE.UU.: “A Bayesian approach for the segmentation of series corrupted by a functional part”.

XIV CLAPEM, December 2016, San José, Costa Rica: “A Bayesian approach for the segmentation of series corrupted by a functional part”.

Joint Statistical Meeting 2016, August 2016, Chicago, EE.UU.: “Predicting Pregnancy Outcomes Using Longitudinal Information: A P-splines Mixed-effects Model Approach”.

XLII Jornadas Nacionales de Estadística, October 2015, Concepción, Chile: “Bayesian regression analysis of data with random effects covariates from nonlinear longitudinal measurements”.

V Encuentro Iberoamericano de Biometría, September 2015, Bilbao, España: “Function selection in mixed effects models using L1-penalization”.

XXVIIth International Biometric Conference, July 2014, Florencia, Italia: “Classification of longitudinal data through a semiparametric mixed-effects model based on lasso-type estimators”.

Joint Statistical Meeting 2014, July 2014, Boston, EE.UU.: “Linear Semiparametric Estimation Procedure for the Segmentation of Multiple Series Based on a Lasso Strategy”.

Joint Statistical Meetings 2012, July 2012, San Diego, EE.UU.: “Lasso-type estimators for Semiparametric Nonlinear Mixed-Effects Models Estimation”.

Charla CIMFAV, July 2012, Valparaíso, Chile: “Estimation in Semiparametric Nonlinear Mixed-Effects Models Using  $\ell_1$ -Penalization”.

1ère Rencontre Francochilienne de Statistique, April 2012, París, Francia: “Estimation in semiparametric nonlinear mixed-effects model using l1-penalization”.

Joint Statistical Meetings 2011, August 2011, Miami, EE.UU: “Classification of Multivariate Longitudinal Data Through a Semiparametric Mixed-Effects Model with Nonstandard Assumptions”.

Seminaire de l’AgroParisTech, June 2011, París, Francia: “Estimation in Nonlinear Mixed-Effects Models Using Heavy-Tailed Distributions”.

Seminaire MODAL’X, Université Paris Ouest Nanterre, June 2011, París, Francia: “Estimation in Nonlinear Mixed-Effects Models Using Heavy-Tailed Distributions”.

Seminario del Departamento de Estadística, PUC, March 2011, Santiago, Chile: “New challenges in Semiparametric Nonlinear Mixed-Effects Models Estimation”.

XXVth International Biometric Conference, December 2010, Florianópolis, Brasil: “New approaches for semiparametric estimation in nonlinear mixed-effects models”.

Joint Statistical Meetings 2010, August 2010, Vancouver, Canadá: “Exact estimation procedures in a spatial mixed-effects probit model with binary outcomes”.

X Jornadas Nacionales de Bioestadística, January 2010, Santiago, Chile: “Exact estimation procedures in complex mixed-effects models: interacting R with WinBUGS”.

Simposio en Modelamiento Estadístico 2009, December 2009, Valparaíso, Chile: “Flexible nonlinear mixed-effects models”.

XVII Semana de la Estadística, October 2009, Valparaíso, Chile: “Algoritmos EM estocásticos y aplicaciones en biometría”.

Seminario Departamento de Estadística, Universidad de Concepción, September 2009, Concepción, Chile: “Modelos mixtos no lineales flexibles”.

VII Escuela de Invierno de Análisis Estocástico y Aplicaciones Valparaíso, July 2009, Valparaíso, Chile: “Modelos mixtos no lineales flexibles”.

Departamento de Estatística, Universidade Estadual de Campinas, May 2009, Campinas, Brasil: “Estimation in nonlinear mixed effects models using heavy-tailed distributions”.

Seminario CIMFAV, Universidad de Valparaíso, April 2009, Valparaíso, Chile: “Extensiones al modelo con efectos mixtos no lineal”.

XI Escola de Modelos de Regresso, March 2009, Recife, Brasil: “Estimation in non-linear mixed-effects model under heavy-tailed distributions using a stochastic approximation of EM algorithm”.

LXXVIII Encuentro Anual Sociedad de Matemática de Chile, November 2008, Valparaíso, Chile: “Modelos Mixtos no lineales con distribuciones de colas pesadas: una aplicación a la farmacocinética”.

VII CLATSE (Congreso Latinoamericano de Sociedades de Estadística), October 2008, Montevideo, Uruguay: “Estimación en modelos no lineales con efectos mixtos usando distribuciones con colas pesadas”.

Joint Statistical Meetings 2008, August 2008, Denver, EE.UU.: “A Robust Nonlinear Mixed-Effects Model Using the SAEM Algorithm”.

XXIV International Biometric Conference, July 2008, Dublin, Irlanda: “Estimation in the probit normal model for binary outcomes using the SAEM algorithm”.

VIII Jornadas Nacionales de Bioestadística, January 2008, Santiago, Chile: “Estimación en el modelo probit normal para datos binarios usando el algoritmo SAEM”.

Seminario Universidad Santa María, November 2007, Valparaíso, Chile: “Una versión estocástica del algoritmo EM. Aplicaciones a los modelos con efectos mixtos”.

Conferencia, XXXIV Jornadas Nacionales de estadística, October 2007, Antofagasta, Chile: “Métodos de estimación exactos para modelos lineales generalizados con efectos mixtos: Aplicaciones al modelo Probit para datos binarios”.

Seminario ANESTOC, Pontificia Universidad Católica de Chile, August 2007, Santiago, Chile: “Una versión estocástica del algoritmo EM. Aplicaciones a los modelos con efectos mixtos”.

Seminario CIMFAV, Universidad de Valparaíso, July 2007, Valparaíso, Chile: “Métodos de estimación exactos para modelos lineales generalizados con efectos mixtos: Aplicaciones al modelo Probit para datos binarios”.

XXXIX Journées de Statistique de la SFDS, June 2007, Orléans, Francia: “Estimation dans un modèle mixte probit normal pour données binaires en utilisant l’algorithme SAEM et ses dérivées”.

Working group MONOLIX, February 2007, Orsay, Francia: “L’algorithme SAEM pour les modèles linéaires généralisés mixtes”.

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## MEMBER OF SCIENTIFIC SOCIETIES

Institute of Mathematical Statistics (IMS), International Society for Bayesian Analysis (ISBA), Sociedad Chilena de Estadística (SOCHE) y International Biometrics Society (IBS).

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## AWARDS AND SCHOLARSHIP

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|-----------|-------------------------------------------------------------------------------|
| 2004-2006 | Doctoral Scholarship, granted by CONICYT and French Embassy in Chile, CHILE.  |
| 2000-2001 | M.S. Scholarship, granted by DIPUC, Pontificia Universidad Católica de Chile. |

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## COMPUTATIONAL SKILLS

Matlab, R, SAS, WinBUGS y S-Plus.

Valparaíso, 2026